INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE CAMPUS
(Affiliated to UEM Kolkata and Approved by AICTE)

Degree:**B.C.A**, Stream:**B.C.A**

End Semester Examination 2023 - First Semester Examination,December-2023

Paper Code: BCACC101

Paper Name: Introduction to C Programming

Full Marks:100 Duration:3 Hours

Part - A
Attempt 10 questions
Each question carries 2 Marks (2 X 10)

1. How many types of loops are there in C programming? 2

or

1. Describe the software with different types. 2

2. State the basic structure of a C program with an example. 2

or

2. Explain the utility main() in a C program? 2

3. Describe the int, long int, short int. 2

or

3. Explain auto keyword. 2

4. Which standard library string function converts a string into lower case? 2

or

4. Explain actual parameters in user defined function. 2

5. What is the difference between a null pointer and an uninitialized pointer? 2

or

5. What is size of union in C? 2

6. Define pointer to pointer. 2

or

6. Describe the statement int *p. 2

7. Discuss nested if-else statement. 2

or

7. Define string with example. 2

8. Why decision making is important in programming? 2

or

8. What is branching? 2

9. Why repetition is important in programming? 2

or

9. Discuss when array is advantageous? 2

10. Define relational operator. 2

or

10. What are the purposes of having return type of the function? 2

Part - B
Attempt 6 questions
Each question carries 5 Marks (5 X 6)

11. Explain relational operator. 5

or

11. Create an application using C to find highest element present in a user given unique data set. 5

12. "Array takes contagious memory allocation"---Explain with a proper diagram. 5

or

12. A 5 digit positive number is taken as input. Develop a program to find sum of the digits of the number using recursion. 5

13. What are user defined data types? How it is achievable in C? 5

or

13. Illustrate the difference between structure and union. 5

14. Briefly explain five types of variables in C. 5

or

14. Explain increment and decrement operators with examples. 5

15. Differentiate between call by value and call by address. 5

or

15. Develop a C program to obtain factors of a number using functions. 5

16. Discuss advantages of having structure in C programming. 5

or

16. Write a program in C to find the maximum number among three numbers using a pointer. 5

Part - C
Attempt 5 questions
Each question carries 10 Marks (10 X 5)

17. Create a C application to read 10 elements from user and find out a specific element from a data set of 10 integer array. 10

or

17. What are format specifiers in C? Explain with example any five format specifiers. 10

18. Develop a C program to calculate electric bill amount using following conditions: - 10

if amount is within 800 then no tax required. If amount is more than or equal to 800 but less than 1200 then 10 %(on bill amt.) of tax will be added to the amount. If amount is more than or equal to 1200 but less than 2000 then 15 %(on bill amt.) of tax will be added to the amount. If amount is more than 1999 then 22% of tax will be charged.

or

18. Explain different memory allocation and management functions in C with proper examples. 10

19. Explain various pre-processor directives with example present in C. 10

or

19. Design a C program to take total marks of a student & determine his/her division using following conditions: - if marks is >=600 then 1st divisions or if marks <600 but marks>=450 then 2nd division or marks<450 but marks>=300 then 3rd divisions. Otherwise fail. 10

20. Write an algorithm to find out the area and perimeter of a circle. 10

or

20. There is a structure called employee that holds information like employee code, name and year of joining. Write a program to create an array of structures and enter some data into it. Develop a C application that displays the names of those employees who joined after 2020. 10

21. Illustrate different functions with its applications used in memory management in C programming language. 10

or

21. Create a C based application to find out 2nd minimum number present in array of 10 integer. 10